

Medical Robotics

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Lecture 1

Organization and Overview

2nd Semester, Academic Year 2024–2025

23 February – 29 May 2026 · 60 hours · 6 ECTS

Lecture notes compiled from slides and lecture transcript.
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1. Course Goals, Methodology, and Prerequisites

1.1. Goals

The course aims to provide an overview of the use of robotic technologies in healthcare, with **special emphasis on robot-assisted surgery**.

1.2. Methodology

The course follows a structured methodology, which is also the guiding thread for the entire program:

- Associate **robotic functions** to **clinical needs**
- Identify the **robot architecture**, **control modality**, **sensors** and **actuators** that best suit the identified needs
- Analyse examples of robotic systems *in clinical use* and of *prototype systems*

1.3. Prerequisites

A good knowledge of robot kinematics and control is assumed.

2. Course Organization

2.1. Schedule

The course runs in the **2nd semester**: 23 February – 29 May 2026, for a total of **60 hours (6 ECTS credits)**.

Day	Time	Room	Students
Tuesday	14:00–16:00	A6, via Ariosto 25	All
Wednesday	10:00–12:00	A6, via Ariosto 25	All
Friday	08:00–10:00	A4, via Ariosto 25	All
Friday	10:00–12:00	A4, via Ariosto 25	MBIR only (likely ends mid-April)

2.2. Contacts and Resources

- **Office hours**: Wednesday 12:30–14:30 (send a message at least one day before)
- **E-learning**: <https://elearning.uniroma1.it/enrol/index.php?id=7465>
- **Email**: marilena.vendittelli@uniroma1.it

3. Exam Structure

There are two **alternative** modalities for passing the exam.

3.1. Option 1 – Project

- Usually requires programming
- Group work, groups of **3–4 students**
- **Necessary condition** for project assignment: 2 homeworks assigned during the course must be completed with a grade of **at least B (≥ 20)**
- Homeworks are **individual** work (not group)

- **Grading weight:** project 90%, homeworks 10%

Project presentation requirements:

- Written report with selected results
- PDF, software, videos, and presentation — all shared in a Drive folder
- **20-minute live presentation** (PowerPoint or other)

Examples of last-year projects are available in the e-learning environment.

Practical advice: Mixed groups from different curricula and cultural backgrounds are strongly encouraged. An important objective of the course is learning to communicate with people from different professional backgrounds — a skill that is essential in the medical robotics field, where engineers must translate clinical needs into technological solutions.

3.2. Option 2 – Written Exam + Oral Discussion

- No homework required
- The oral discussion can involve any topic in the program
- Examples of written exam texts are available in the e-learning environment

4. Dr. Robot: Science Fiction or Reality?

The course opens by asking: how far is science fiction from the real state of medical robotics? Fictional robots from films such as *Humans*, *Robot & Frank*, *Big Hero 6*, *Elysium*, *Ender's Game*, and *Dr. Easy* are depicted as performing surgery, assisting people, advising, healing, and providing physical strength (exoskeletons). The course examines which of these roles are already a reality and how far research has progressed toward the others.

4.1. A Robotic Tool for Each User Category

There exist three main categories of people who interact with medical robots:

User Category	Robotic Tools
Clinicians / Surgeons	Robots for surgery, exploration, diagnosis, and therapy delivery
Therapists	Robots and mechatronic tools for therapy and training
Patients	Robots and machines that improve quality of life of elderly and disabled people, mainly by increasing their personal autonomy

Note the distinction between **robots** and **mechatronic tools** in the therapist category: these are close relatives sharing actuators and sensors, but robots have a degree of autonomy that mechatronic tools may not. The boundary is a spectrum rather than a sharp line.

5. Surgery Transformation Through Technology Innovation

The transformation of surgery is driven by innovation along **four dimensions**:

- **Materials:** from rigid to soft
- **Size:** from macro to micro

- **Access:** from open to minimally invasive to ingestible/injectable
- **Dexterity, accuracy, autonomy:** from no autonomy to high-level autonomy, increase in internal degrees of freedom, shared control with the operator

5.1. The Spectrum of Surgical Approaches

The slide illustrates a progression of surgical paradigms, from oldest to most advanced:

Paradigm	Description
Open	Large incision; direct manual access to the operative field
Minimally invasive	Small incisions; indirect access via instruments inserted through trocars
Endoluminal	Access via natural body cavities (e.g., mouth) using flexible instruments
Flexible	Flexible robotic tools that can navigate complex anatomical paths
Micro	Microrobots operating at sub-millimeter scales
Self-assembling	Robots that assemble inside the body from smaller components (research frontier)
Nano	Nanoscale devices for targeted drug delivery and cellular-level intervention

This progression is not purely sequential — multiple paradigms coexist today, and research advances across all of them simultaneously. Importantly, this transformation is not due to robotics alone: it results from a convergence of innovation in materials science, imaging, computing, miniaturization, and AI.

6. Milestones in Surgery Innovation

The following timeline summarizes the key milestones presented in the course slides.

6.1. 1985 – PUMA 560 in Neurosurgery

The **PUMA 560** (Unimation), an industrial robot, was the first robot used in a medical procedure: CT-guided stereotactic brain biopsy in neurosurgery (Kwoh et al., 22 patients). Neurosurgery was the entry point because the task was essentially **needle placement with high positional accuracy** — exactly what an industrial robot does well.

Workflow: preoperative CT scan → 3D brain model reconstruction → biopsy target planning → robot positions needle guide with millimetric accuracy → surgeon manually inserts the needle.

The PUMA was also used for **transurethral resection of the prostate** at Imperial College (around the same era).

6.2. 1989 – Speedy in Neurosurgery

The **Speedy** robot, developed by Lavallée, Benabid et al., was used in neurosurgery and treated **hundreds of patients**. This represented an important step beyond the initial proof of concept, demonstrating the clinical viability of robotic assistance in stereotactic procedures.

6.3. 1992 – ROBODOC in Orthopedic Surgery

ROBODOC (Integrated Surgical Systems) was the first **purpose-built** surgical robot, designed specifically for **total hip replacement**. Over 70 robots were deployed, treating over **10,000 patients**. The robot milled the femoral canal to precise geometries required for implant fitting — a task of machining rigid surfaces directly analogous to industrial CNC machining.

6.4. 1994 – AESOP in MIS

AESOP (Automated Endoscopic System for Optimal Positioning, Computer Motion) was the first robot used in **minimally invasive surgery (MIS)**. It was a **voice-commanded endoscope holder**: the robot held and moved the laparoscopic camera in response to surgeon voice commands, freeing the surgical team from manual camera control. Over 400 units were deployed.

6.5. 1997 – Da Vinci (Intuitive Surgical)

The **Da Vinci** system (Intuitive Surgical) was introduced as a complete **teleoperated MIS system**, consisting of a surgeon console and patient-side manipulator arms. The installed base exceeded 850 systems. Da Vinci dominates the surgical robot market to this day.

6.6. 1998 – Zeus (Computer Motion)

Zeus (Computer Motion) was a competing complete MIS system. It was involved in the landmark **Lindbergh Operation** (2001): the first transatlantic teleoperated surgery, in which a surgeon in **New York** performed a laparoscopic cholecystectomy (gallbladder removal) on a patient in **Strasbourg**, demonstrating the technical feasibility of remote surgery over long distances.

6.7. 2003 – MARS: Bone-Mounted Robot

MARS (Mazor Technologies) was introduced in 2003 as a **bone-mounted** robot for spinal surgery. Its key design choice was small size and a constrained workspace, making it **intrinsically safer** than large-arm robots. This reflected a broader design debate: task-specific small robots vs. general-purpose large industrial manipulators adapted for surgery.

6.8. 2008 – First NOTES Procedure

NOTES (Natural Orifice Transluminal Endoscopic Surgery) exploits natural body orifices (mouth, urethra, vagina) to access internal organs with no external incisions. Internal incisions heal faster and carry lower infection risk than external ones.

IRCAD (Institut de Recherche contre les Cancers de l'Appareil Digestif, Strasbourg) is famous for "Operation Anubis". This was the first "totally" scarless (pure NOTES) cholecystectomy (gallbladder removal) performed on a human.

The route was Transvaginal. It was the first time a major organ was removed in a human using only a natural orifice, without any external abdominal incisions. It was a massive media and medical milestone.

6.9. 2019 onwards – New Systems and Growing Competition

- **Da Vinci Single Port**: all instruments through a single incision, typically the umbilicus; used for urological and transoral procedures

- **Senhance** (TransEnterix): a new MIS system offering more cost-effective and flexible robotic assistance
- **Ion** (Intuitive Surgical): a robotic-assisted bronchoscopy platform for lung biopsy (60-subject trial). The system is well-established, with over 900 systems installed globally and more than 140,000 lung biopsies performed in 2025 alone. In late 2025, the FDA cleared significant software updates for Ion that integrate artificial intelligence and advanced imaging. These updates help doctors account for "CT-to-body divergence" (when a lung nodule moves between the time of the scan and the procedure).
- **Hugo** (Medtronic): another competing MIS platform

Feature	da Vinci (Intuitive Surgical)	Hugo RAS (Medtronic)
Configuration	Integrated: A single large cart with 4 fixed arms.	Modular: Individual, wheeled carts for each arm (up to 4).
Console Design	Closed: Immersive "head-in" view; limits peripheral vision.	Open: Surgeon wears 3D glasses; allows easier talk with the team.
Pricing	Premium: Generally \$1.5M – \$2.5M.	Competitive: Typically ~30% lower than da Vinci Xi.
Current Status	Market leader with 8,000+ units globally.	Expanding; FDA cleared (Dec 2025) for urology.
Tech Edge	da Vinci 5 features "Force Feedback" (feeling tissue resistance).	Touch Surgery: Deep AI integration for video analytics/training.

Figure 1: daVinci vs Hugo main features

- General trend: **more cost-effective and flexible robot-assisted surgical systems for MIS**, as Da Vinci patents expire and the market opens to competition

7. Current Challenges and Future Developments

7.1. Current Challenges

Despite significant progress, medical robotic systems face important practical challenges that limit wider adoption:

- **Cost:** high costs of installation, maintenance, and per-procedure use (the Da Vinci system costs \$1–2M to purchase, plus significant maintenance and disposable instrument costs)
- **Size and footprint:** large systems occupy significant operating room space and can alter surgical workflow
- **Increased operational times and cognitive load:** robot-assisted procedures can be longer than conventional ones; **situation awareness** — the surgeon's real-time understanding of what is happening in and around the patient — is critical and can be reduced when operating through a console
- **The long way to certification:** regulatory approval for medical robots is lengthy and complex (no specific regulatory framework exists for medical robots; existing frameworks for medical devices and industrial robots only partially apply)

7.2. Future Technologies

Three converging technological directions are expected to mitigate these challenges:

- **VR/AR:** virtual and augmented reality for education, training, surgical planning, and intraoperative navigation
- **Micro/nano robotics:** miniaturized devices for local drug delivery and microsurgery, enabling intervention at scales impossible with current systems
- **AI-powered systems:** artificial intelligence for surgical guidance, decision support, skill assessment, and increasing levels of autonomous assistance

8. From Industrial to Medical Robots: What Do They Share?

A central question of the course: medical robotics did not emerge in isolation — it grew directly from industrial robotics research and practice.

8.1. Shared Architecture

Industrial and medical robots share the same fundamental architecture:

- Mechanical structure (links, joints, end-effector)
- Sensors (proprioceptive and exteroceptive)
- Actuators (typically electric motors, but not limited to)
- Control system

The analytical tools — kinematics, dynamics, control theory — developed for industrial robots apply directly to medical robots. This is why prerequisites for the course include solid knowledge of robot kinematics and control.

8.2. What Is Specific to Medical Robots

While the architecture is shared, medical robots face additional constraints and requirements:

- **Size:** instruments must possibly pass through small incisions or natural orifices
- **Sterilization:** all patient-contact components must be sterilizable or disposable
- **Biocompatibility:** materials must be safe for contact with human tissue
- **Safety:** the robot operates on a human patient — safety requirements are uniquely stringent
- **Regulatory compliance:** must satisfy medical device regulations
- **Control:** the presence of the human in the loop (both patient and surgeon) introduces specific control challenges

9. Course Program Overview

The full program is available on the e-learning platform. As summarized in the slides, the course covers:

1. **Organization and overview** (this lecture): course structure, exam modalities, overview of the field
2. **Evolution of surgical robots:** detailed historical development and current state of the art

3. **Methodology:** from clinical need to robotic function to robot design
4. **Application domains:** orthopedics, neurosurgery, MIS, microsurgery, rehabilitation, and more
5. **Surgical systems in clinical use:** primarily da Vinci and competitors, but not limited to; classification and schematic design
6. **Control modalities:** teleoperation, autonomous control, cooperative/shared control
7. **Sensing and state estimation:** sensors in medical robotics; observability; state estimation when sensing is constrained
8. **Haptics:** force/tactile/vibration/temperature feedback; haptic interfaces and rendering; sensory substitution
9. **Simulation:** surgical training; data generation for AI; real-time deformable tissue simulation (problem not yet effectively solved)
10. **Safety and regulation:** safety in human–robot interaction; medical device regulation; certification
11. **AI in medical robotics:** cognitive augmentation; current applications and future directions
12. **Exoskeletons and biomechanics of walking** (shorter coverage this year)

10. Bibliography

10.1. Medical Robotics Specific References

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Note on course material: Last-year material available in the e-learning environment will be updated during the course. Specific bibliographic references are included at the end of each lecture's slide set.